THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue _____
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Student Number |__|__|__|__|__|__|__|__|
Family Name _____
First Name _____

School of Electrical Engineering & Computer Science

Semester Two Examinations, 2024

ELEC7309 Power System Planning and Reliability

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•No written or printed material permitted
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

2 x 14-Page Answer Booklet
1 x 1cm x 1cm Graph Paper

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

There are two Sections in this exam (PART - A and PART - B). Answer each section (PART - A and PART - B) in a separate Booklet. This final exam accounts for 60% or 60 marks out of 100 marks (full mark of this course).

For Examiner Use Only

Question	Mark

Total _____

PART A: POWER SYSTEM PLANNING
ANSWER ALL TWO QUESTIONS FROM PART A

Question A1 **(Total marks 15)**

(a) An expansion project has two plans – Plan A and Plan B – as shown in Table 1 (unit: k$). Compare these two plans and comment on which plan is better in terms of the least cost. Use constant annuity for the analysis.

(7 marks)

Table 1 for Question A1 (a): Details of Plan A and Plan B

Project	Plan A	Plan B
Investment	3500	5000
Residual value (RV)	0	1500
Lifetime (years)	6	8
Annual cost	800	500
Discount rate (%)	5.5	5.5

(b) For three generation technology types A, B, and C, with details in Table 2, assume an interest rate of 6% and the possibility of choosing any generation capacity of the above-mentioned three technologies (A, B and C).

Table 2 for Question A1 (b)

Technology type	Investment cost ($/kW)	Plant life (Years)	Fuel cost ($/MWh)
A	2500	30	15
B	1000	20	40
C	500	15	70

(1) Derive the equations of annual cost as a function of capacity factor for each of the technologies using the data given in Table 2. **(3 marks)**

(2) Find the optimal generation type and capacity to supply the demand characterised by the Load Duration Curve shown in Figure 1. **(5 marks)**

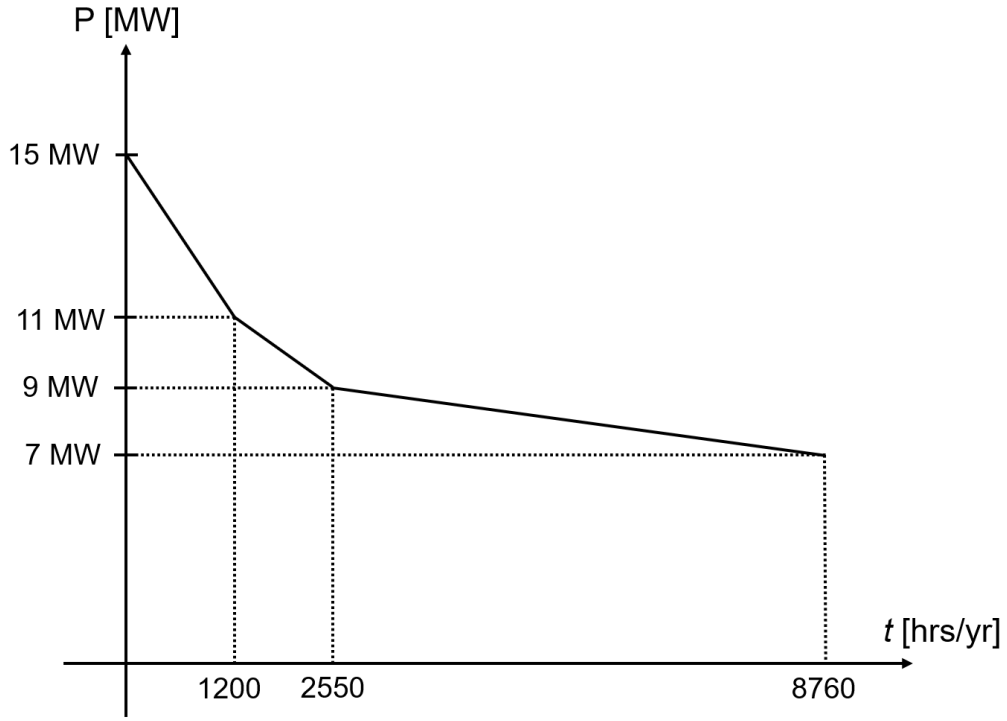


Figure 1 for Question A1 (b): Load duration curve

Question A2 **(Total marks 15)**

(a) Examine a peer-to-peer marketplace where 8 consumers and 9 prosumers participate. The functioning of this marketplace entails giving priority to consumers with the lowest demand and prosumers with the highest surplus. However, if multiple consumers and prosumers bid for the same amount, they are served according to their bidding sequence (first bid first serve). A mid-market rule is employed to establish the price for peer-to-peer transactions. The sequence of bids for various consumers are detailed in Table 3, while those for prosumers are presented in Table 4.

It's important to note that during this period, electricity procurement from the grid costs 27 cents per kWh, and the feed-in tariff rate stands at 9 cents per kWh. It's assumed that all consumers have batteries, and they meet their 50% demand using their batteries before participating in the P2P market. For example, if a consumer's demand is 6 kWh for the considered timeslot, it consumes 3 kWh from its own battery and then participate in the P2P market to meet the rest 3 kWh of demand. No prosumers have batteries.

Compute the earnings and expenses of individual customers based on their available supply and demand during the chosen trading period.

(7 marks)

Table 3 for Question A2 (a): Demand of various consumers.

Bidding sequence	1	2	3	4	5	8	6	8
Consumer	A	B	C	D	E	F	G	H
Demand (kWh)	8	6	8	8	4	12	13	11

Table 4 for Question A2 (a): Supply from various prosumers.

Bidding sequence	1	2	3	4	5	6	7	8	9
Prosumer	M	N	O	P	Q	R	S	T	U
Surplus (kWh)	6	5	4	3.5	4.5	3.5	4	4	5.5

(b) Consider the system shown in Figure 2. Assume that the transfer capacity between the two regions (from Region A to Region B) is 350 MW. Both regions have the same marginal costs of generation indicated by σ_A and σ_B for region A and region B respectively. Here:

$$\sigma_A = 15 + 0.03\, P_A \text{ (in \$/MWh)}$$

$$\sigma_B = 17 + 0.04\, P_B \text{ (in \$/MWh)}$$

and P_A and P_B are in MW.

(1) Calculate the price of electricity in both regions. **(2 marks)**
(2) Calculate the total cost of generation production per hour. **(2 marks)**
(3) Calculate the cost per hour of electricity for customers in both regions. **(2 marks)**
(4) Calculate the profits made by the generators per hour in both regions. **(2 marks)**

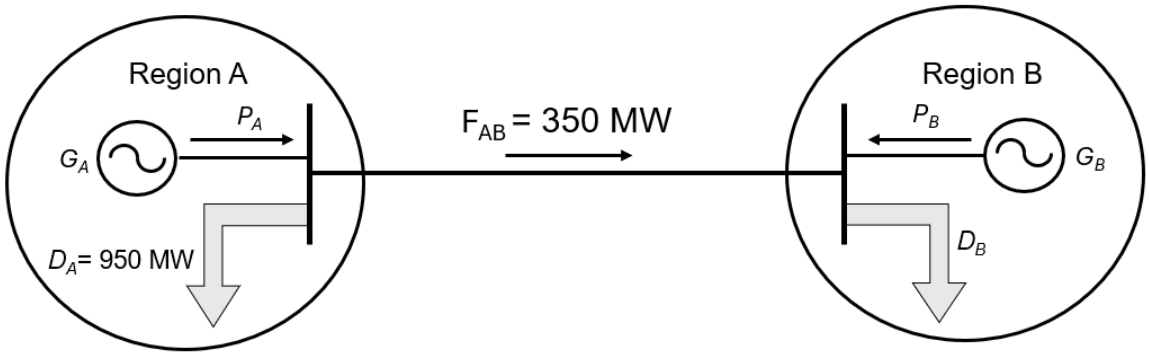


Figure 2 for Question A2 (b)

PART B: POWER SYSTEM RELIABILITY QUESTIONS START ON THE NEXT PAGE

PART B: POWER SYSTEM RELIABILITY
ANSWER ALL TWO QUESTIONS FROM PART B – USE A NEW ANSWER BOOK

Question B1 **(Total marks 15)**

A power system contains three generating units as shown in Table 5. The system's daily peak load over 20 days is recorded in Table 6.

Table 5 for Question B1: Generation Details

Generator	Capacity (MW)	Failure rate (failure/day)	Repair rate (repair/day)
1	25	0.02	0.48
2	25	0.02	0.48
3	50	0.02	0.48

Table 6 for Question B1: Load Profile

Peak load (MW)	Days	Peak load (MW)	Days
60	6	50	5
55	4	45	5

(a) Formulate the capacity outage probability table of the system including both individual outage probability and cumulative outage probability. **(12 marks)**

(b) Calculate the Loss of Load Expectation (LOLE) of the system. (**3 marks)**

Question B2 **(Total marks 15)**

Consider a single line representation of a radial three-phase power system is shown in Figure 3, which contains only one circuit breaker. Assume that any fault, single phase or otherwise, will trip all three phases. Assume that the failure rate of the lines is approximately proportional to their length. For example, the failure rate of the main feeder (sections 1, 2, 3, and 4) is 0.1 f/km/yr and the failure rate of the lateral distribution lines (a, b, c, and d) is 0.3 f/km/yr. Assume that the repair time *r* for each of the main feeder is 3 hours and each of the lateral distribution lines is 2 hours.

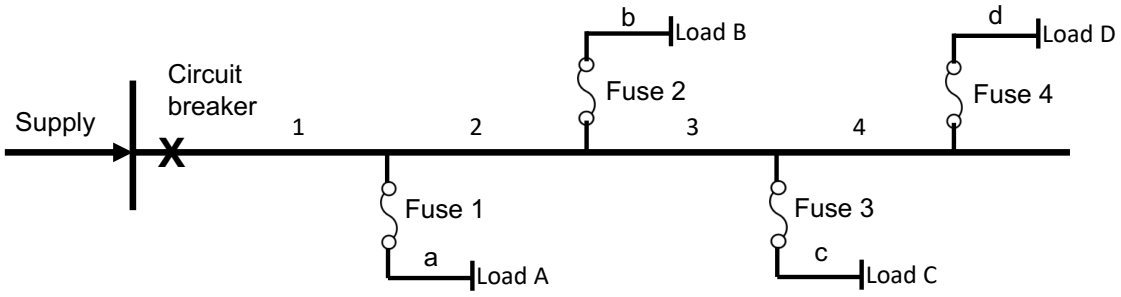


Figure 3 for Question B2

Reliability parameters for the system of Figure 3 are shown in Table 7. Customers and load connected to the system in Figure 3 are shown in Table 8.

Table 7 for Question B2: Reliability Parameters

Component	Length (km)	Component	Length (km)
Section- 1	2	Distributor -a	1
Section- 2	2	Distributor -b	2
Section- 3	1	Distributor -c	1
Section- 4	2	Distributor -d	1

Table 8 for Question B2: Customers and Load Details

Load point	Number of customers	Average load connected (kW)
A	400	2000
B	250	1500
C	300	1800
D	200	1000

Assume that short circuits happen to each component and each failure on the main feeder will trip the main circuit breaker. A fuse is installed at the tee-point in each lateral distribution line which will blow when a short circuit happens on a lateral distributor. On this basis, what would be the reliability indices of the distribution system including SAIFI, SAIDI, CAIDI, ASUI and ENS?

END OF EXAMINATION

FORMULAE NEEDED FOR ANSWERING THE QUESTIONS ARE ON THE FOLLOWING PAGES

FORMULAE – PART A

Variable names and meaning

P_0	Present value
F_n	Value in n years
A	Annuity (constant)
A_n	Annuity in year n
i	Interest rate
B_{jt}	Profit of plan j in year t
C_{jt}	Operation cost of plan j in year t
K_{jt}	Investment of plan j in year t
j	Number of the plan in consideration
n_j	Horizon of study (service/lifetime) of plan j
x_g	Electricity selling price of the grid
x_{FT}	Feed-in-tariff price
x_{MP}	Mid-market price
π	Marginal cost of the generator
P	Power produced by the generator
R	Revenue to the generator
PC	Production cost to the generator
D	Total demand of the customer
J	Cost per hour for customers

Present value

$$P_0 = \frac{F_n}{(1+i)^n}$$

Present Value Factor $(P|F,i,n) = \frac{1}{(1+i)^n}$

Future value

$$F_n = (1+i)^n \,.P_0$$

Future Value Factor $(F|P,i,n) = (1+i)^n$

Constant Annuity

$$F_n = A\frac{(1+i)^n - 1}{i} = A(F|A,i,n)$$

$$A = F_n\frac{i}{(1+i)^n - 1} = F_n(A|F,i,n)$$

$$P_0 = \frac{F_n}{(1+i)^n} = A\frac{(1+i)^n - 1}{i \cdot (1+i)^n} = A\frac{1-(1+i)^{-n}}{i} = A(P|A,i,n)$$

$$A = P_0\frac{i \cdot (1+i)^n}{(1+i)^n - 1} = P_0\frac{i}{1-(1+i)^{-n}} = P_0(A|P,i,n)$$

Capital Recovery Factor

$$CRF_j = \frac{i \cdot (1+i)^{n_j}}{(1+i)^{n_j} - 1} = \frac{i}{1-(1+i)^{-n_j}} = (A|P,i,n_j)$$

Incremental Annuity

$$P_0 = \sum_{k=1}^{n} A_k(1+i)^{-k} = \sum_{k=1}^{n} A_k(P|F,i,k)$$

Net present value

$$NPV_j = \sum_{t=1}^{n_j}\left[B_{jt}(P|F,i,t)\right] - \sum_{t=1}^{n_j}\left[(C_{jt}+K_{jt})(P|F,i,t)\right]$$

Benefit/Cost Ratio

$$BCR_j = \frac{\sum_{t=1}^{n_j}\left[(B_{jt}-C_{jt})(P|F,i,t)\right]}{\sum_{t=1}^{n_j}\left[K_{jt}(P|F,i,t)\right]}$$

Present Value of Costs

$$PVC_j = \sum_{t=1}^{n_j}\left[\left(C_{jt}+K_{jt}\right)(P|F,i,t)\right]$$

Mid-market Price (Price set via mid-market rule)

$$x_{MP} = \frac{x_g + x_{FT}}{2}$$

Capacity Factor (CF)

$$CF = \frac{kWh\ produced\ in\ a\ year}{rated\ power\ (kW) \times 8760h/year}$$

Revenue to generator per hour

$$R(P) = \pi \cdot P$$

Production cost to generator per hour

$$PC(P) = \int_0^{P_B} \pi_B dP$$

Cost per hour for customers

$$J = \pi \cdot D$$

FORMULAE -PART B

General

Failure rate/Forced Outage Rate: λ

Repair Rate: μ

Mean Time Before Failure (MTBF) $\frac{1}{\lambda}$

Mean Time To Repair (MTTR): $r = \frac{1}{\mu}$

For Generation and Composite Power System Reliability Assessment

Availability: $A = \frac{\mu}{\mu+\lambda}$

Unavailability: $U = \frac{\lambda}{\mu+\lambda}$

For Distribution Reliability Assessment

Availability: $A = 1$

Unavailability: $B = \lambda \cdot r = \frac{\lambda}{\mu}$

Common Customer Reliability Indices

System Average Interruption Frequency Index

$$\text{SAIFI} = \frac{\text{total number of customer interruptions}}{\text{total number of customers served}} = \frac{\sum N_i \cdot F_i}{\sum N_i}$$

System Average Interruption Duration Index

$$\text{SAIDI} = \frac{\text{sum of all customer interruption duration}}{\text{total number of customers served}} = \frac{\sum N_i \cdot Q_i}{\sum N_i}$$

Customer Average Interruption Frequency Index

$$\text{CAIFI} = \frac{\text{total number of customer interruptions}}{\text{total number of customers affected}} = \frac{\sum N_i \cdot F_i}{\sum N_i}$$

Customer Average Interruption Duration Index

$$\text{CAIDI} = \frac{\text{sum of all customer interruption duration}}{\text{total number of customer interruptions}} = \frac{\sum N_i \cdot Q_i}{\sum N_i \cdot F_i}$$

Average Service Availability Index

$$\text{ASAI} = \frac{\text{customer hours of available service}}{\text{customer hours demanded}} = \frac{\sum N_i \cdot (1 - Q_i)}{\sum N_i}$$

Average Service Unavailability Index $\text{ASUI} = 1 - \text{ASAI}$

Energy Not Supplied

$$\text{ENS} = \sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}$$

Average Energy Not Supplied

$$\text{AENS} = \frac{\text{total energy not supplied}}{\text{numbers of customers served}} = \frac{\sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}}{\sum N_i}$$

Average Curtailment Index

$$\text{ACCI} = \frac{\text{total energy not supplied}}{\text{number of customers affected}} = \frac{\sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}}{\sum N_i}$$